Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/12/2008
PI Name:	Wong, Joyce Ph.D.		
Project Title:	Novel Microstructures for Vascular Tissue Engineering		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Boston University		
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City:	Boston	**State:**	MA
Zip Code:	02215-2407	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Desai, Tejal (Boston University)		
Grant/Contract No.:	NNJ04HC73G		
Performance Goal No.:			
Performance Goal Text:			
	The major goal of this research project is to tissue-engineer small diameter blood vessel substitutes. Such tissue-engineered blood vessels can also serve as tissue models for studying the effects of microgravity environmental changes on tissue adaptation and to assess potential therapeutic strategies. Requirements for a blood vessel substitute include sufficient burst pressure, viscoelastic mechanical properties, vasoactivity, and nonthrombogenicity. Native blood vessels have evolved a sophisticated structure and hierarchical organization of cell types and matrix molecules in order to satisfy these requirements. Our central hypothesis is that the physiological spatial arrangement of cells and extracellular matrix in an engineered blood vessel will affect the viscoelastic mechanical properties, which in turn, will affect vascular smooth muscle and endothelial cell response. This hypothesis will be tested by answering questions directly posed in Specific Aims 1 and 2		

Task Description:	in which the effects of two main biomaterial scaffold properties on tissue function will be explored: (1) microstructure and (2) mechanics. Our strategy of fabricating novel scaffolds with defined microstructure and mechanical properties will help to reveal the effects of specific ultrastructural and mechanical factors on cell response and tissue development. We will use this approach to develop a new generation of biomaterials for vascular tissue engineering that will provide an in vitro model tissue in which parameters can be varied systematically to investigate cell mechanics, tissue development, cell-substrate and cell-cell interactions in controlled microgravity environments in order to attain a better understanding of effects of microgravity on human physiology. Furthermore, we recognize that these tissues may also be used as potential blood vessel replacements that are in high demand and are an integral part of more complex organs and tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will have direct impact because we are developing tissue engineered blood vessels. We will be able to use these blood vessels as model systems to test the effects of zero-gravity and pharmacological effects. Unfortunately, this program was ended prematurely (by one year before its original end date), so we were not able to continue the work in the bioreactor which would have simulated the zero-gravity environment.
Task Progress:	We have published two peer-reviewed papers in 2005, 3 peer-reviewed papers in 2006, 1 peer-reviewed paper in 2007, 2 peer-reviewed papers in 2008, and another paper in review, and two papers in preparation. This makes a total of 8 published papers since 2005 and potentially an additional 3 papers. We have made significant progress in investigating the role of microstructure on vascular smooth muscle cells, specifically studying the role of substrate microtopography combined with mechanical cyclic strain on cell orientation. We also developed novel methodologies to self-assemble biomacromolecules such as collagen that are critical components of blood vessels. We also showed that we could stack several layers of microtopographically patterned thin scaffold films and keep the cells viable in this configuration, a critical step in forming the blood vessels.
Bibliography Type:	Description: (Last Updated: 01/09/2014)
Articles in Peer-reviewed Journals	Houtchens GR, Foster MD, Desai TA, Morgan EF, Wong JY. "Combined effects of microtopography and cyclic strain on vascular smooth muscle cell orientation." J Biomech. 2008;41(4):762-9. PMID: 18222460 , Jan-2008
Articles in Peer-reviewed Journals	Köster S, Leach JB, Struth B, Pfohl T, Wong JY. "Visualization of flow-aligned type I collagen self-assembly in tunable pH gradients." Langmuir. 2007 Jan 16;23(2):357-9. PMID: 17209575 , Jan-2007
Articles in Peer-reviewed Journals	Isenberg BC, Wong JY. "Building structure into engineered tissues. Review." Materials Today. 2006 Dec;9(12): 54-60. http://dx.doi.org/10.1016/S1369-7021(06)71743-6 , Dec-2006
Articles in Peer-reviewed Journals	Sarkar S, Dadhania M, Rourke P, Desai TA, Wong JY. "Vascular tissue engineering: microtextured scaffold templates to control organization of vascular smooth muscle cells and extracellular matrix." Acta Biomater. 2005 Jan;1(1):93-100. PMID: 16701783 , Jan-2005
Articles in Peer-reviewed Journals	Sarkar S, Lee GY, Wong JY, Desai TA. "Development and characterization of a porous micro-patterned scaffold for vascular tissue engineering applications." Biomaterials. 2006 Sep;27(27):4775-82. PMID: 16725195 , Sep-2006
Articles in Peer-reviewed Journals	Brown XQ, Ookawa K, Wong JY. "Evaluation of polydimethylsiloxane scaffolds with physiologically-relevant elastic moduli: interplay of substrate mechanics and surface chemistry effects on vascular smooth muscle cell response." Biomaterials. 2005 Jun;26(16):3123-9. PMID: 15603807 , Jun-2005
Articles in Peer-reviewed Journals	Jacot JG, Dianis S, Schnall J, Wong JY. "A simple microindentation technique for mapping the microscale compliance of soft hydrated materials and tissues." J Biomed Mater Res A. 2006 Dec 1;79(3):485-94. PMID: 16779854 , Dec-2006
Articles in Peer-reviewed Journals	Sarkar S, Isenberg BC, Hodis, E, Leach JB, Desai TA, Wong JY. "Fabrication of a layered microstructured polycaprolactone construct for 3-D tissue engineering." Journal of Biomaterials Science, Polymer Edition. 2008;19(10):1347-62. PubMed PMID: 18854127 , Sep-2008
Dissertations and Theses	Sarkar S. "Fabrication of a layered micro-structured tissue construct for vascular tissue engineering applications." MS Thesis, Boston University, 2006. , May-2006